INCH-POUND

MIL-C-3098/95C
5 May 1993
SUPERSEDING
MIL-C-3098/95B
23 June 1975

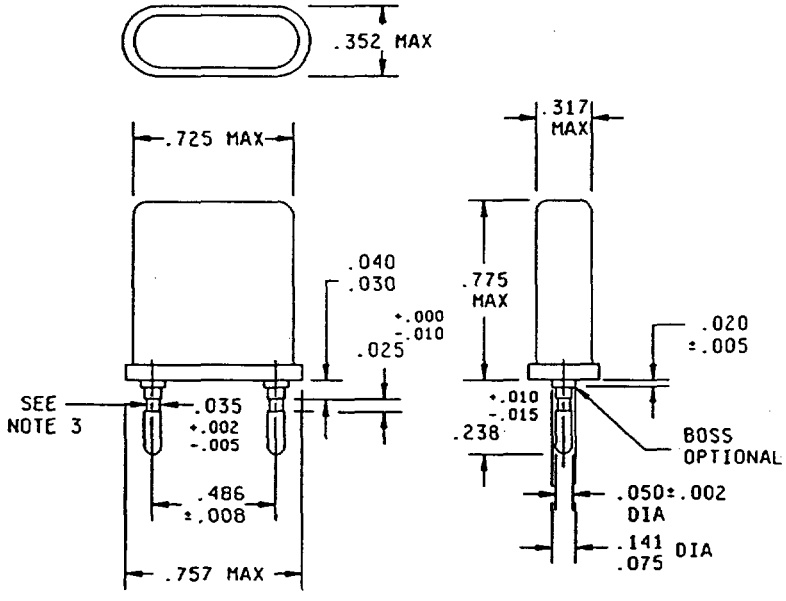
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR119/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; antiresonance.




| Inches | mm |
|---|---|
| .002 | .05 |
| .005 | .13 |
| .008 | .20 |
| .010 | .25 |
| .015 | .38 |
| .020 | .51 |
| .025 | .64 |
| .030 | .76 |
| .035 | .89 |
| .040 | 1.02 |
| .050 | 1.27 |
| .075 | 1.91 |
| .141 | 3.58 |
| .238 | 6.05 |
| .317 | 8.05 |
| .352 | 8.94 |
| .486 | 12.34 |
| .725 | 18.42 |
| .757 | 19.23 |
| .775 | 19.69 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR119/U.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-6/U (MIL-H-10056/2). 1/

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 to 20 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

| Frequency range, inclusive | Maximum resistance |
|---|---|
| MHz | Ohms |
| 0.8 to 0.85 | 620 |
| 0.85+ to 0.9 | 600 |
| 0.9+ to 1 | 570 |
| 1+ to 1.12 | 540 |
| 1.12+ to 1.25 | 490 |
| 1.25+ to 1.37 | 450 |
| 1.37+ to 1.5 | 410 |
| 1.5+ to 1.62 | 370 |
| 1.62+ to 1.75 | 330 |
| 1.75+ to 1.87 | 300 |
| 1.87+ to 2.0 | 290 |
| 2.0+ to 2.12 | 270 |
| 2.12+ to 2.25 | 240 |
| 2.25+ to 2.6 | 190 |
| 2.6+ to 3 | 150 |
| 3+ to 3.4 | 110 |
| 3.4+ to 3.75 | 90 |
| 3.75+ to 4 | 75 |
| 4+ to 5 | 60 |
| 5+ to 7 | 35 |
| 7+ to 10 | 24 |
| 10+ to 15 | 22 |
| 15+ to 20 | 20 |

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level: 0.8 to 10.0 MHz = 10.0 ±2.0 mW.
10.0+ to 20.0 MHz = 5.0 ±1.0 mW.

Capacitance, shunt: 7.0 pF maximum.

Antiresonance load capacitance: 30.0 ±0.5 pF 2/

| Shock: (specified pulse) | 0.8 to 2.0 MHz | 2.0+ to 20.0 MHz |
|---|---|---|
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |

1/ The socket designed to mate with crystal holder HC-6/U is designated in MIL-S-12883/4 (Sockets, Crystal, Top and Bottom Mounting, 2 Contacts). For Navy use, other sockets listed in MIL-STD-242 may be used.
2/ This crystal unit is similar to Crystal Type CR18/U, except for the 30.0 ±0.5 pF antiresonance load capacitance.



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| | | |
|---|---|---|
| Vibration: Method 201 of MIL-STD-202. | 0.8 to 2.0 MHz | 2.0+ to 20.0 MHz |
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |
| Thermal shock: | 0.8 to 2.0 MHz | 2.0+ to 20.0 MHz |
| Frequency change permitted: | ±10 ppm | ±5 ppm |
| Equivalent resistance change permitted: | ±15 percent | ±10 percent |

Aging:

Frequency change permitted: ±5 ppm

NOTE: NATO preferred type NXT2/A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

## CONCLUDING MATERIAL

Custodians:

Army - ER
Navy - EC
Air Force - 85

Review activities:

Army - AR, MI, SM

User activities:

Navy - AS, MC, SH
Air Force - 17, 19, 84

Preparing activity:

Army - ER

Agent:

DLA - ES

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